

Paper 1 - Set 08 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	F	Indices and surds
2	A	Indices and surds
3	D	Quadratic functions
4	C	Quadratic functions
5	B	Simultaneous equations and inequalities
6	D	Probability
7	C	Pythagoras, mensuration and 3D
8	B	Functions and transformations
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11	E	Polynomials, factor and remainder theorems
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18	B	Straight lines
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Worked solutions

Question 1 IND-01 · Indices and surds · spec MM1.1

Simplify $\frac{(2x^3)^4 \times 8x^{-5}}{4x^2}$.

Answer: F

$(2x^3)^4 = 16x^{12}$, so the numerator is $16x^{12} \times 8x^{-5} = 128x^7$. Dividing by $4x^2$ gives $32x^5$.

Full method - how and why

A pure index-laws question. Nothing here needs thought once you treat the numbers and the powers of x as two separate bookkeeping jobs.

1 Deal with the bracket first: $(2x^3)^4 = 2^4 x^{12} = 16x^{12}$.

why: A power outside a bracket multiplies **every** factor inside, so the 2 is raised to the 4th power as well as the x^3 . This is the step people skip.

2 Multiply the numerator: $16x^{12} \times 8x^{-5} = 128x^7$.

why: Multiplying powers of the same base adds the indices: $12 + (-5) = 7$. The numbers just multiply normally.

3 Divide by $4x^2$: $\frac{128x^7}{4x^2} = 32x^5$.

why: Dividing subtracts indices: $7 - 2 = 5$, and $128 \div 4 = 32$.

Where it goes wrong

- Forgetting to raise the 2 to the 4th power gives $2x^{12}$ at step 1 and the answer $4x^5$.
- Subtracting instead of adding the -5 gives x^9 and the answer $32x^9$ - which is offered.

Takeaway Handle the number part and the index part separately, and always distribute an outside power over everything in the bracket.

Question 2 IND-03 · Indices and surds · spec MM1.1

Solve $4^{x+1} = 8^{2x-3}$.

Answer: A

Write both sides as powers of 2: $2^{2x+2} = 2^{6x-9}$. So $2x + 2 = 6x - 9$, giving $4x = 11$ and $x = \frac{11}{4}$.

Full method - how and why

An exponential equation where both sides can be written as powers of the **same** base. Once the bases match, the indices must be equal.

1 Spot that $4 = 2^2$ and $8 = 2^3$.

why: Both numbers are powers of 2, so the whole equation can be rewritten in base 2 and the exponentials compared directly.

2 Rewrite: $4^{x+1} = 2^{2(x+1)} = 2^{2x+2}$ and $8^{2x-3} = 2^{3(2x-3)} = 2^{6x-9}$.

why: $(a^m)^n = a^{mn}$: the outer index multiplies the whole inner index, so brackets are essential.

3 Equate the indices: $2x + 2 = 6x - 9$, so $4x = 11$ and $x = \frac{11}{4}$.

why: $2^u = 2^v$ forces $u = v$ because 2^x is a one-to-one function (it never takes the same value twice).

Where it goes wrong

- Multiplying out as $3 \times 2x - 3$ instead of $3(2x - 3)$ loses the bracket and gives a different linear equation.
- Writing $8 = 2^4$ (a slip) gives $x = \frac{7}{2}$, which is offered.

Takeaway Whenever an equation has different bases, look for a common base first; comparing indices is then legitimate because exponentials are one-to-one.

Question 3 QUA-10 · Quadratic functions · spec MM1.3

The quadratic $x^2 + px + q$ has minimum value -4 and one of its roots is 0 . Which of the following could be the quadratic?

Answer: D

Minimum -4 means $x^2 + px + q = (x - h)^2 - 4$, with roots $h \pm 2$. One root is 0 , so $h = \pm 2$: the quadratic is $x^2 - 4x$ or $x^2 + 4x$. Only $x^2 - 4x$ is offered.

Full method - how and why

Two pieces of information about a quadratic - its minimum and one root - pin the whole thing down. Working in completed-square form makes the roots visible.

- 1 Minimum -4 and leading coefficient 1 means $x^2 + px + q = (x - h)^2 - 4$ for some h .
why: A monic quadratic with minimum m must be $(x - h)^2 + m$; only the position h is unknown.
- 2 Its roots satisfy $(x - h)^2 = 4$, i.e. $x = h \pm 2$.
why: Solve the completed-square form directly rather than expanding.
- 3 One root is 0 , so $h + 2 = 0$ or $h - 2 = 0$: $h = -2$ or $h = 2$.
why: Both cases are legitimate; the options decide which one is wanted.
- 4 That gives $x^2 + 4x$ or $x^2 - 4x$; the list offers $x^2 - 4x$.
why: Check it: $x^2 - 4x = (x - 2)^2 - 4$, minimum -4 , roots 0 and 4 .

Where it goes wrong

- Picking $x^2 - 2x - 3$: its minimum is -4 but its roots are -1 and 3 , so 0 is not a root.
- Checking only the minimum (several options have minimum -4) or only the roots.

Takeaway Test every stated condition - an option that satisfies one of them is designed to catch you.

Question 4 QUA-16 · Quadratic functions · spec MM1.3

Given that $x^2 + y^2 = 10$ and $x + y = 4$, find the value of xy .

Answer: C

$(x + y)^2 = x^2 + y^2 + 2xy$, so $16 = 10 + 2xy$ and $xy = 3$.

Full method - how and why

A symmetric pair of equations. Expanding $(x + y)^2$ links the two given quantities to xy directly.

- 1 Write the identity $(x + y)^2 = x^2 + y^2 + 2xy$.
why: This is the only identity that connects a sum, a sum of squares and a product.
- 2 Substitute the given values: $4^2 = 10 + 2xy$.
why: $x + y = 4$ and $x^2 + y^2 = 10$ are both given, so only xy is unknown.
- 3 Solve: $16 - 10 = 2xy \Rightarrow xy = 3$.
why: Halve at the end - forgetting to divide by 2 gives 6 , which is offered.

Where it goes wrong

- Answering 6 by omitting the factor 2 in $2xy$.
- Trying to solve for x and y individually (they are 1 and 3) - slower and unnecessary.

Takeaway $(x + y)^2 = x^2 + y^2 + 2xy$ converts between the three symmetric quantities; recognise which two you are given.

Question 5 SIM-17 · Simultaneous equations and inequalities · spec MM1.4

Two numbers have sum 10 and the sum of their squares is 58. What is the positive difference between them?

Answer: B

$(x - y)^2 = 2(x^2 + y^2) - (x + y)^2 = 116 - 100 = 16$, so the difference is 4 (the numbers are 3 and 7).

Full method - how and why

Symmetric information again: a sum and a sum of squares. The identity for $(x - y)^2$ gets the difference without finding the numbers.

1 Use $(x - y)^2 = x^2 + y^2 - 2xy$ and $(x + y)^2 = x^2 + y^2 + 2xy$.

why: Adding these gives $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$ - the relation that links all three given quantities.

2 Rearrange: $(x - y)^2 = 2(x^2 + y^2) - (x + y)^2$.

why: This avoids having to find xy as an intermediate step.

3 Substitute: $2(58) - 10^2 = 116 - 100 = 16$.

why: Careful: it is $2(x^2 + y^2)$, not $x^2 + y^2$.

4 Take the positive square root: the difference is 4 (the numbers are 3 and 7).

why: The question asks for the **positive** difference, so take +4.

Where it goes wrong

- Forgetting the factor 2 and getting $(x - y)^2 = -42$, which is impossible.
- Giving 16 (the square of the difference) as the answer.

Takeaway $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$: any two of sum, difference and sum-of-squares give the third.

Question 6 PRO-09 · Probability · spec M7.4

The probabilities that a train is early, on time or late are 0.1, x and $2x$ respectively. Find the probability that the train is not late.

Answer: D

$0.1 + 3x = 1 \Rightarrow x = 0.3$; not late = $0.1 + 0.3 = 0.4$.

Full method - how and why

The probabilities of an exhaustive set of outcomes sum to 1. Use that to find x first.

1 Early, on time, late are the only possibilities, so $0.1 + x + 2x = 1$.

why: Exhaustive and mutually exclusive outcomes have total probability 1.

2 Solve: $3x = 0.9$, so $x = 0.3$ and late = 0.6.

why: Subtract 0.1, then divide by 3.

3 Not late = early + on time = $0.1 + 0.3 = 0.4$.

why: Or use the complement: $1 - 0.6 = 0.4$ - quicker and a useful check.

Where it goes wrong

- Stopping at $x = 0.3$ and answering that, or answering 0.6, the probability of being late.
- Forgetting 'early' counts as 'not late'.

Takeaway Probabilities of all outcomes sum to 1; 'not late' is the complement of 'late'.

Question 7 MEN-16 · Pythagoras, mensuration and 3D · spec M5.18

From the top of a cliff 40 m high, the angle of depression of a boat is 30° . How far is the boat from the foot of the cliff?

Answer: C

$$\tan 30^\circ = \frac{40}{d} \Rightarrow d = \frac{40}{1/\sqrt{3}} = 40\sqrt{3}.$$

Full method - how and why

An angle of depression from the top equals the angle of elevation from the boat. Then it is right-angled trigonometry.

- 1 Draw the triangle: the cliff is vertical (40 m), the sea horizontal, and the angle at the boat is 30° .
why: The angle of depression from the cliff top equals the angle of elevation from the boat (alternate angles with the horizontal).
- 2 Relative to the 30° angle, the cliff is opposite and the distance d is adjacent: $\tan 30^\circ = \frac{40}{d}$.
why: $\tan = \frac{\text{opposite}}{\text{adjacent}}$.
- 3 Solve: $d = \frac{40}{\tan 30^\circ} = \frac{40}{1/\sqrt{3}} = 40\sqrt{3}$ m.
why: Dividing by $\frac{1}{\sqrt{3}}$ multiplies by $\sqrt{3}$.

Where it goes wrong

- Multiplying by $\tan 30^\circ$ instead of dividing, giving $\frac{40}{\sqrt{3}}$ - an offered option.
- Placing the 30° angle at the top of the cliff inside the triangle, which would swap opposite and adjacent.

Takeaway Depression from the top = elevation from the bottom; a shallow angle means a large horizontal distance.

Question 8 FUN-05 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ passes through (4, 6). Which point must lie on the graph of $y = 3f(2x) - 1$?

Answer: B

$f(2x)$ halves x -coordinates: (2, 6). Then $3f - 1$: $y = 18 - 1 = 17$. Point (2, 17).

Full method - how and why

Track a single known point through the transformations, remembering that inside changes act inversely on x .

- 1 Start with the known point (4, 6) on $y = f(x)$.
why: One point is all you are given, so the answer must be its image.
- 2 $f(2x)$ squashes horizontally by factor $\frac{1}{2}$: the point with x -coordinate 4 moves to $x = 2$, since $f(2 \times 2) = f(4) = 6$.
why: To get the same output from $f(2x)$ you need half the input - this is the inverse action of an inside change.
- 3 Multiply by 3: $y = 3 \times 6 = 18$. Then subtract 1: $y = 17$.
why: Outside changes act directly on the output, in the order written.
- 4 The point is (2, 17).
why: Check: $3f(2 \times 2) - 1 = 3(6) - 1 = 17$ ✓.

Where it goes wrong

- Doubling the x -coordinate to 8 instead of halving it.
- Forgetting the -1 and answering (2, 18).

Takeaway Substitute the point back into the new equation to check: $f(2x)$ at $x = 2$ really is $f(4)$.

Question 9 SIM-14 · Simultaneous equations and inequalities · spec MM1.4

Solve $2x + 3y = 12$ and $x^2 - y^2 = 0$. How many solutions (x, y) are there?

Answer: B

$x^2 = y^2$ means $y = x$ or $y = -x$. $y = x$: $5x = 12$, one solution. $y = -x$: $-x = 12$, one solution. Two in total.

Full method - how and why

$x^2 - y^2 = 0$ factorises into two lines. The system is a line meeting a pair of lines, so count the intersections of each case.

- Factorise: $x^2 - y^2 = (x - y)(x + y) = 0$, so $y = x$ or $y = -x$.
why: A product is zero when either factor is zero - this splits the problem into two straight-line cases.
- Case $y = x$: $2x + 3x = 12 \Rightarrow 5x = 12$, one solution $(\frac{12}{5}, \frac{12}{5})$.
why: Substitute into the linear equation; a linear equation in one unknown has exactly one root.
- Case $y = -x$: $2x - 3x = 12 \Rightarrow -x = 12$, one solution $(-12, 12)$.
why: Same method; the coefficient of x is non-zero so again exactly one solution.
- Total: two solutions.
why: The two cases give different points, so no double counting.

Where it goes wrong

- Taking the square root as $y = x$ only, halving the answer.
- Concluding 'a line meets a quadratic in at most 2 points, so 2' without checking that neither case is parallel or degenerate.

Takeaway $A^2 = B^2$ means $A = B$ or $A = -B$; both branches must be followed.

Question 10 MEN-03 · Pythagoras, mensuration and 3D · spec M5.16

A trapezium has parallel sides of length 8 cm and 14 cm and area 77 cm^2 . Find the perpendicular distance between the parallel sides.

Answer: C

$$\frac{1}{2}(8 + 14)h = 77 \Rightarrow h = 7.$$

Full method - how and why

Trapezium area formula, rearranged for the height.

- Write the formula: area $= \frac{1}{2}(a + b)h$ where a, b are the parallel sides.
why: The height is the **perpendicular** distance between the parallel sides.
- Substitute: $\frac{1}{2}(8 + 14)h = 77$.
why: $\frac{1}{2} \times 22 = 11$.
- Solve: $11h = 77$, so $h = 7$ cm.
why: Divide both sides by 11.

Where it goes wrong

- Forgetting the $\frac{1}{2}$ and answering 3.5 - an offered option.
- Using the difference of the parallel sides instead of the sum.

Takeaway Trapezium: average the parallel sides, then multiply by the perpendicular height.

Question 11 POL-14 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$ is a cubic polynomial with $p(1) = p(2) = p(3) = 0$ and $p(0) = 12$. Find $p(4)$.

Answer: E

$$p(x) = a(x-1)(x-2)(x-3); p(0) = -6a = 12 \Rightarrow a = -2. p(4) = -2 \cdot 3 \cdot 2 \cdot 1 = -12.$$

Full method - how and why

Three roots determine a cubic up to a constant multiple. One extra value fixes the multiple.

- 1 Write $p(x) = a(x-1)(x-2)(x-3)$.

why: Given roots, start from the factorised form; the leading coefficient a is the only freedom left.

- 2 Use $p(0) = 12$: $a(-1)(-2)(-3) = -6a = 12$, so $a = -2$.

why: Three negative factors give a negative product - a very common sign slip.

- 3 Evaluate $p(4) = -2(3)(2)(1) = -12$.

why: Substitute directly into the factorised form; no expansion needed.

Where it goes wrong

- Taking $a = 2$ from $6a = 12$ and answering $+12$.
- Assuming $a = 1$ because the question says 'a cubic' - the leading coefficient is not given.

Takeaway Roots give the shape, one extra point gives the scale; keep everything in factorised form.

Question 12 ALG-07 · GCSE algebra · spec M4.6

Simplify $\frac{3}{x+2} - \frac{2}{x-1}$.

Answer: D

$$\frac{3(x-1) - 2(x+2)}{(x+2)(x-1)} = \frac{x-7}{(x+2)(x-1)}.$$

Full method - how and why

Subtracting algebraic fractions: common denominator, then expand the numerator carefully - especially the minus sign.

- 1 Common denominator: $(x+2)(x-1)$.

why: The denominators share no factor, so their product is the lowest common denominator.

- 2 Rewrite each fraction: $\frac{3(x-1)}{(x+2)(x-1)} - \frac{2(x+2)}{(x+2)(x-1)}$.

why: Multiply each numerator by whatever its denominator was missing.

- 3 Combine and expand: $3x - 3 - (2x + 4) = 3x - 3 - 2x - 4 = x - 7$.

why: The minus sign applies to **both** terms of the second bracket - the classic error.

- 4 So the answer is $\frac{x-7}{(x+2)(x-1)}$.

why: Numerator does not factorise further.

Where it goes wrong

- Getting $x+1$ by only negating the $2x$ - an offered option.
- Subtracting the denominators as well.

Takeaway Bracket the whole second numerator before subtracting, then distribute the minus sign.

Question 13 CIR-07 · Circles and circle theorems · spec MM3.3

$PQRS$ is a cyclic quadrilateral with angle $P = 3x$, angle $R = x + 40^\circ$. Find x .

Answer: C

Opposite angles of a cyclic quadrilateral sum to 180° : $4x + 40 = 180$, $x = 35^\circ$.

Full method - how and why

Cyclic quadrilateral: opposite angles are supplementary. That gives a linear equation immediately.

- 1 Identify the opposite pair: in $PQRS$ the vertices go round in order, so P and R are opposite.

why: Opposite means separated by one vertex on each side - reading the labelling order matters.

- 2 Apply the theorem: $3x + (x + 40) = 180$.

why: Opposite angles of a cyclic quadrilateral sum to 180° .

- 3 Solve: $4x = 140$, so $x = 35^\circ$.

why: Check: angles 105° and 75° sum to 180° ✓.

Where it goes wrong

- Setting the angles equal (that is a parallelogram property, not a cyclic one).
- Using 360° and getting $x = 80^\circ$.

Takeaway Cyclic quadrilateral: opposite angles sum to 180° ; adjacent ones have no such relation.

Question 14 TRG-01 · Trigonometric functions and equations · spec MM4.6

How many solutions does $\sin 2x = \frac{1}{2}$ have for $0 \leq x < 2\pi$?

Answer: C

$2x$ runs over $[0, 4\pi)$, which contains two full periods of $\sin$, each giving two solutions: 4 in total.

Full method - how and why

Counting solutions of $\sin(kx) = c$. Transform the interval for the inside expression and count there.

- 1 Let $u = 2x$. As x runs over $[0, 2\pi)$, u runs over $[0, 4\pi)$.

why: Multiplying the variable by 2 doubles the range of the inside expression - this is the whole trick.

- 2 $[0, 4\pi)$ contains exactly two full periods of $\sin$.

why: $\sin$ has period 2π , so a 4π -long interval holds two complete waves.

- 3 In each period, $\sin u = \frac{1}{2}$ has two solutions (one on the way up, one on the way down).

why: A horizontal line strictly between -1 and 1 cuts each wave twice.

- 4 Total: $2 \times 2 = 4$ solutions.

why: Each u gives exactly one x via $x = u/2$, so no solutions are lost or duplicated.

Where it goes wrong

- Answering 2 by counting solutions for x as if the 2 were not there.
- Answering 8 by doubling twice.

Takeaway For $\sin(kx) = c$ on an interval of length L , the inside runs over length kL - count solutions there.

Question 15 LIN-06 · Straight lines · spec MM3.1

The perpendicular bisector of the line segment joining (1, 5) and (7, 1) has equation

Answer: A

Midpoint (4, 3); gradient of segment $\frac{-4}{6} = -\frac{2}{3}$, so the bisector has gradient $\frac{3}{2}$:
 $y - 3 = \frac{3}{2}(x - 4) \Rightarrow 2y - 6 = 3x - 12 \Rightarrow 3x - 2y = 6$.

Full method - how and why

Perpendicular bisector: through the midpoint, perpendicular to the segment. Two small calculations, then the point-gradient form.

1 Midpoint: $(\frac{1+7}{2}, \frac{5+1}{2}) = (4, 3)$.

why: The bisector must pass through the middle of the segment.

2 Gradient of the segment: $\frac{1-5}{7-1} = -\frac{4}{6} = -\frac{2}{3}$.

why: Keep the order of subtraction consistent in numerator and denominator.

3 Perpendicular gradient: $\frac{3}{2}$.

why: Negative reciprocal of $-\frac{2}{3}$.

4 Line through (4, 3): $y - 3 = \frac{3}{2}(x - 4) \Rightarrow 2y - 6 = 3x - 12 \Rightarrow 3x - 2y = 6$.

why: Multiply by 2 to clear the fraction, then collect.

Where it goes wrong

- Rearranging to $3x - 2y = 18$ by a sign slip on the constant.
- Using the segment's gradient instead of the perpendicular one.

Takeaway Perpendicular bisector = midpoint + negative reciprocal gradient; it is the locus of points equidistant from the two ends.

Question 16 LOG-05 · Exponentials and logarithms · spec MM5.2

Solve $\log_3(x + 6) + \log_3(x - 2) = 2$.

Answer: C

$(x + 6)(x - 2) = 9 \Rightarrow x^2 + 4x - 12 = 0 \Rightarrow (x + 7)(x - 3) = 0$. $x = -7$ makes both logs undefined, so $x = 3$ only.

Full method - how and why

Combine the logs, convert to an equation, then **check the domain** - that is what this question is really testing.

1 Combine: $\log_3[(x + 6)(x - 2)] = 2$.

why: Sum of logs is the log of the product.

2 Convert to exponential form: $(x + 6)(x - 2) = 3^2 = 9$.

why: $\log_a N = c$ means $N = a^c$.

3 Expand and solve: $x^2 + 4x - 12 = 9 \Rightarrow x^2 + 4x - 21 = 0 \Rightarrow (x + 7)(x - 3) = 0$, so $x = -7$ or $x = 3$.

why: Ordinary quadratic work.

4 Check both in the **original** equation: $x = 3$ gives $\log_3 9 + \log_3 1 \checkmark$; $x = -7$ gives $\log_3(-1)$, which is undefined.

So $x = 3$ only.

why: Logarithms of negative numbers (or zero) do not exist, so roots that break the original expression must be rejected.

Where it goes wrong

- Giving both roots - the option ' $x = -7$ or $x = 3$ ' exists precisely to catch this.
- Checking only one of the two logarithms.

Takeaway Always substitute your solutions back: combining logs can create solutions the original equation never had.

Question 17 LOG-08 · Exponentials and logarithms · spec MM5.3

Given that $5^x = 2$ and $2^y = 125$, find the value of xy .

Answer: B

$$5^{xy} = (5^x)^y = 2^y = 125 = 5^3, \text{ so } xy = 3.$$

Full method - how and why

Two exponential facts that chain together. Raising the first to the power y links them.

- 1 Raise the first equation to the power y : $(5^x)^y = 2^y$, i.e. $5^{xy} = 2^y$.

why: $(a^x)^y = a^{xy}$ - this is what connects the two statements.

- 2 Substitute the second fact: $2^y = 125$.

why: Given directly.

- 3 Write 125 as a power of 5: $5^{xy} = 5^3$.

why: Both sides are now powers of 5.

- 4 Equate the indices: $xy = 3$.

why: Valid because 5^x is one-to-one.

Where it goes wrong

- Trying to compute x and y separately as logarithms and getting stuck without a calculator.
- Multiplying the two equations to get $10^{x+y} = 250$, which leads nowhere.

Takeaway To find a **product** of exponents, raise one equation to the other's exponent and chain the results.

Question 18 LIN-04 · Straight lines · spec MM3.1

The lines $y = 2x + 1$, $y = -x + 7$ and the x -axis enclose a triangle. Find its area.

Answer: B

$$\text{Intersection: } 2x + 1 = 7 - x \Rightarrow x = 2, y = 5. \text{ } x\text{-intercepts: } -\frac{1}{2} \text{ and } 7, \text{ base} = \frac{15}{2}. \text{ Area} = \frac{1}{2} \cdot \frac{15}{2} \cdot 5 = \frac{75}{4}.$$

Full method - how and why

A triangle bounded by two lines and an axis. Find the three vertices, then use the axis segment as the base.

- 1 Find where the two lines meet: $2x + 1 = -x + 7 \Rightarrow 3x = 6 \Rightarrow x = 2, y = 5$. Apex (2, 5).

why: Equate the two expressions for y .

- 2 Find the x -intercepts: $2x + 1 = 0$ gives $x = -\frac{1}{2}$; $-x + 7 = 0$ gives $x = 7$.

why: The third side is along the x -axis, so the other two vertices are the x -intercepts.

- 3 Base length: from $-\frac{1}{2}$ to 7 is $7 - (-\frac{1}{2}) = \frac{15}{2}$.

why: Subtract, do not add; the negative intercept makes the base longer, not shorter.

- 4 Height is the y -coordinate of the apex, 5: area $= \frac{1}{2} \times \frac{15}{2} \times 5 = \frac{75}{4}$.

why: With a horizontal base, the height is just the apex's distance from the axis.

Where it goes wrong

- Using base $7 - \frac{1}{2} = \frac{13}{2}$ by mishandling the negative intercept.
- Using the apex x -coordinate 2 as the height.

Takeaway Put the base on an axis whenever you can: the height is then simply a coordinate.

Question 19 CIR-09 · Circles and circle theorems · spec MM3.2

Find the length of the tangent from the point (7, 1) to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$.

Answer: B

Centre (1, 2), radius 3. Distance from (7, 1) to centre: $\sqrt{36 + 1} = \sqrt{37}$. Tangent length = $\sqrt{37 - 9} = \sqrt{28}$.

Full method - how and why

The tangent, radius and line to the external point form a right-angled triangle. Pythagoras gives the tangent length.

- 1 Put the circle in centre-radius form: $(x - 1)^2 + (y - 2)^2 = 4 + 1 + 4 = 9$, so centre (1, 2), radius 3.

why: Complete the square in x and y ; the constant -4 moves across.

- 2 Distance from (7, 1) to the centre: $\sqrt{(7 - 1)^2 + (1 - 2)^2} = \sqrt{36 + 1} = \sqrt{37}$.

why: This is the hypotenuse of the right-angled triangle (the tangent meets the radius at 90°).

- 3 Tangent length = $\sqrt{PC^2 - r^2} = \sqrt{37 - 9} = \sqrt{28}$.

why: Subtract the squares, not the lengths - the right angle is at the point of contact.

Where it goes wrong

- Answering $\sqrt{37}$ (that is the distance to the centre) or $\sqrt{37} - 3$.
- Getting $r = 9$ instead of 3 from the completed square.

Takeaway Tangent length from P : $\sqrt{PC^2 - r^2}$, because tangent $\perp$ radius makes a right-angled triangle.

Question 20 BIN-14 · Binomial expansion · spec MM2.4

How many terms in the expansion of $(x + y + z)^4$ (after collecting like terms) are there?

Answer: B

Terms are $x^a y^b z^c$ with $a + b + c = 4$, $a, b, c \geq 0$. Counting: for $a = 0$ there are 5 choices of (b, c) , $a = 1$: 4, ..., $a = 4$:

1. Total $5 + 4 + 3 + 2 + 1 = 15$.

Full method - how and why

A trinomial expansion. Counting the distinct terms is a counting problem: how many ways can the powers add to 4?

- 1 Every term has the form $x^a y^b z^c$ with $a + b + c = 4$ and $a, b, c \geq 0$.

why: Each of the four brackets contributes exactly one of x , y or z , so the powers add to 4.

- 2 Count systematically by the value of a : if $a = 0$ then $b + c = 4$, giving 5 pairs; $a = 1$ gives 4; $a = 2$ gives 3; $a = 3$ gives 2; $a = 4$ gives 1.

why: For a fixed a , the number of (b, c) pairs is $(4 - a) + 1$.

- 3 Total: $5 + 4 + 3 + 2 + 1 = 15$.

why: A triangular number - the standard result $\binom{n+2}{2}$ for three variables, here $\binom{6}{2} = 15$.

Where it goes wrong

- Answering $3^4 = 81$ - that counts the terms **before** collecting like terms.
- Answering 5 by counting only the powers of x .

Takeaway Distinct terms of $(x + y + z)^n$ correspond to solutions of $a + b + c = n$ in non-negative integers: $\binom{n+2}{2}$ of them.